

My approach can i purely detect convergence/divergence of a conversation. cause my current steps is to write up the process of my approach with ABI_S15 so i can really understand it

(Write up to follow on the next page)

Section 1: Data and Embedding Foundations (Plain Version)

For ABI_S15, I started by breaking the transcript down into the smallest possible units: sentences. Each sentence had a timestamp (when it was said), a speaker label (who said it), and sometimes a codebook tag (like “decision,” “explanation,” or “new idea”). This gave me a timeline of the conversation at the sentence level.

Then I turned each sentence into a vector — a mathematical representation of its meaning — using a pretrained language model. This meant I could compare any two sentences not just by their words, but by whether they expressed similar ideas. To measure similarity, I used a score between 0 and 1, where 1 means “these two sentences are saying basically the same thing.” This set up the foundation for detecting when people were aligning in real time.

Section 2: Detecting Convergence Bursts (Plain Version)

Once I could measure how similar sentences were, the next step was figuring out *when* similar things were being said close together in time. To do this, I looked at short time windows around each point in the conversation — for example, a ± 45 second span. Inside each window, I checked all the pairs of sentences said by different people within a short gap.

If a lot of those pairs were highly similar, I called that a convergence burst. To avoid mistaking one person repeating themselves for group agreement, I required at least three different speakers to be part of the burst. This way, a flagged burst meant: multiple people, within about a minute, all saying things that were semantically close. Those flagged windows are the “candidate convergence moments” I can then analyze more closely.

Section 3: Overlay with Codes and Topics (Plain Version)

To interpret the bursts, I layered them on top of two other sources of information: the human-coded labels and the actual technical terms from the transcript.

If a burst overlapped with a coded *decision*, that was strong evidence it was real convergence. If it only overlapped with *explanations* or *new ideas*, then it might just be exploratory. At the same time, I pulled out technical keywords — like “photoacoustic” or “two-photon” — to ground the bursts in actual scientific content.

By plotting bursts, codes, and keywords together, I could see whether clusters of similar speech actually aligned with important discussion points. For example, in the 14–15 minute section, a

burst lined up with a decision code and with key topic terms. That combination suggested the group wasn't just repeating themselves — they were actually converging on a choice.

Section 4: Transcript Validation and Examples (Plain Version)

Finally, I checked the raw transcript for each flagged burst. This step was important because not every burst meant real convergence. For example, the 913–918 second burst looked like convergence computationally, but in the transcript the team didn't reach a decision — they were circling ideas without resolving them. By contrast, in the 14–15 minute burst, the transcript showed genuine alignment around a technical method, confirming the computational flag.

